

UNDERSTANDING CAVITATION IN PROCESS PUMPS

HYDRAULIC TROUBLESHOOTING GUIDE

A field-oriented way to separate cavitation symptoms from other hydraulic or mechanical problems, identify the operating condition that triggers them, and verify the correction.

PRIMARY AUDIENCE	REFERENCE STATUS
Reliability engineers / Operators / Maintenance teams	Illustrative project concept

FIELD SEQUENCE

ENGINEERING LENS	WHAT TO RESOLVE
01 / Recognize the pattern	Listen for a crackling or unstable hydraulic sound and look for vibration, fluctuating flow, loss of head, or a changing motor load. Compare the symptom with the exact flow, temperature, and suction state.
02 / Frame the NPSH question	Compare available suction head with the pump requirement across the real operating envelope. Treat tank level, vapor pressure, upstream losses, temperature, and speed as variables.
03 / Check the suction path	Inspect valve position, strainer condition, line size, air ingress, blocked vents, pipe support, and upstream equipment. A changed process arrangement can remove the original margin.
04 / Correct the cause	Reduce avoidable suction loss, restore liquid level or temperature conditions, change the operating point, or revise the equipment arrangement. Record which variable changed and why.
05 / Verify and monitor	Repeat the measurement at the triggering duty, confirm stable readings, check for damage, and add a practical operating limit or inspection point to the maintenance record.

UNDERSTANDING CAVITATION IN PROCESS PUMPS / HANDOVER RECORD

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A useful field guide ends with evidence that the system is ready for the next team. Capture the records below in the project handover pack and keep exceptions visible until they are closed.

RECOMMENDED RECORDS

01	Symptom and operating-state log
02	Available versus required NPSH review
03	Suction path inspection record
04	Corrective action and verification reading
05	Updated operating limit or monitoring point

REFERENCE NOTE

This guide supports early planning and field conversations. It does not replace the approved drawings, equipment instructions, site procedures, inspection plan, or project acceptance criteria.